

Topic: Design a Maze + Write the Solution · **Course:** Maze Madness · **Level:** Extension (Advanced) ·

Time: about 60 minutes

This time you are the maze maker. Design one maze that **stretches through 6 zones**, wire it with pistons and doors, then write the **solution code** that drives the agent to the end.

Open **M002 EXT 3 — Cube World** and build inside it. Drive the agent with chat: **l** turns left, **r** turns right, **run** starts your solver, **rl** teleports the agent back to you.

Your maze must include at least:

- **2 pistons** (each opens a wall or path when powered)
- **2 doors** (the agent opens them with `agent.interact(...)`)
- **4 AND conditions** in your solution (`... and ...`)
- **1 OR condition** in your solution (`... or ...`)

Tick off each part as you go.

1 • Design Your 6 Zones

Your maze runs through **6 zones** in a row. Give each zone one job: a turn, a piston, a door, or a junction that checks two things.

Sketch your 6 zones. Write each zone number and what makes it tricky.

Where go your 2 pistons? Which zones, and what does each open?

Where go your 2 doors? What is behind each one?

2 · Build the Maze

Build it in the world. Walk it yourself first (no code) to check a person can reach the end, then tick each part as you place it.

- ☐ Zone 1 built
- ☐ Zone 2 built
- ☐ Zone 3 built
- ☐ Zone 4 built
- ☐ Zone 5 built
- ☐ Zone 6 built
- ☐ Piston #1 placed and tested
- ☐ Piston #2 placed and tested
- ☐ Door #1 placed
- ☐ Door #2 placed
- ☐ A clear START block and a clear GOAL block

What broke while you built it? What did you change?


```
while True
```

your maze.

```
while True:
    agent.move(FORWARD, 1)

    # a junction – turn only when BOTH things are true (AND)
    if agent.detect(BLOCK, LEFT) and agent.detect(REDSTONE, AHEAD):
        agent.turn(LEFT_TURN)

    # a door OR a piston gate – handle either one (OR)
    elif agent.detect(BLOCK, AHEAD) or agent.detect(REDSTONE, RIGHT):
        agent.interact(AHEAD)
        agent.move(FORWARD, 1)
```

if

elif

#

AND and each OR with # OR .

4 · Count Your Logic

Go back through your solution and **count**. Write the number, then copy one real line as proof.

How many AND conditions? (need at least 4)

Copy one AND line:

How many OR conditions? (need at least 1)

Copy your OR line:

Why did that spot need OR instead of AND? What two things is it checking for?

5 · Spot and Fix the Bugs

Read each, say what goes wrong, then write the fix.

```
# Should turn left only when a wall is on the left AND redstone is ahead.  
if agent.detect(BLOCK, LEFT) or agent.detect(REDSTONE, AHEAD):  
    agent.turn(LEFT_TURN)
```

What is wrong, and what is the fixed line?


```
# Should open a door ahead, then step through it.  
elif agent.detect(BLOCK, AHEAD):  
    agent.move(FORWARD, 1)  
    agent.interact(AHEAD)
```

What is wrong, and what is the fixed code?

6 · Show Your Work 📷 🎥

Type `run` and watch the agent. It will get stuck somewhere — that is normal. Find where it stops, fix that part of your code or your maze, and run it again. Use `rl` to send it back between tries. Keep going until the agent reaches the **GOAL** all by itself.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.